

AP Chem summer homework

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2 Math problems

- a. 1.81×10^6
- b. 1.43×10^6
- c. 2.482×10^{-7}
- d. -1.738×10^{-7}
- e. 6.2×10^8
- f. 2×10^{-3}
- g. 5
- h. 0.3
- i. 2×10^{-5}
- j. $\pm 16/15$
- k. 4×10^{-5}
- l. 4
- m. -10.68
- n.

$$x^2 + 0.1x = 2 \times 10^{-8}$$
$$x^2 + 0.1x - 2 \times 10^{-8} = 0$$
$$x = \boxed{2 \times 10^{-7}}$$

o.

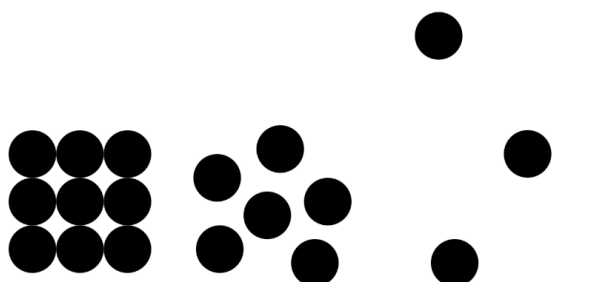
$$x + y = 3$$
$$x - y = 9$$
$$2x = 12$$
$$\boxed{x = 6}$$

3 Significant figures

- a. 5
- b. 3
- c. 4
- d. 3

- e. 5
- f. 1
- g. 1
- h. 6
- i. 5
- j. 6

4 Chapter 1



solid	liquid	gas
• not compressible • fixed shape • fixed volume	• slightly compressible • no fixed shape • fixed volume	• compressible • no fixed shape • no fixed volume

- a.
- b. iii
- c. 19%
- d. Dimensional analysis-
 - i.

$$0.069 \text{ nm} = 6.9 \times 10^{-9} \text{ cm}$$

$$V = \frac{4}{3}\pi(6.9 \times 10^{-9})^3 = 1.38 \times 10^{-24} \text{ cm}^3$$

$$\frac{3.35 \times 10^{-23} \text{ g}}{1.38 \times 10^{-24} \text{ cm}^3} = 24.3 \text{ g/cm}^3$$

- ii. 10^3 ; 10^2 ; 10^{-3} ; 10^{-6} ; 10^{-9}

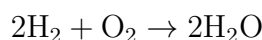
- iii.

$$\frac{330 \text{ minutes}}{1 \text{ day}} \cdot \frac{1 \text{ hour}}{60 \text{ minutes}} = 5.5 \text{ hours/day}$$

- e. Element: oxygen; compound: water; homogeneous mixture: dish soap; heterogeneous mixture: oil in water; solution: saltwater

5 Chapter 2

- a. The mass number is simply the sum of protons plus neutrons in an atom, while average atomic mass is the average of all the masses of all isotopes of an element.
- b. Mass can neither be created nor destroyed
- c.



- d. Molar mass of K_2SO_4 : 174.26 g/mol

$$\text{K}_2—2 \cdot 39.1 = 78.2 \text{ g/mol}$$

$$\text{S}—32.06 \text{ g/mol}$$

$$\text{O}_4—4 \cdot 16 = 64 \text{ g/mol}$$

Assuming 1 mol, K makes up 78.2 of 174.26 grams

$$\frac{78.2}{174.26} \times 100\% = \boxed{44.88\%}$$

- e. Nucleus: center of the atom consisting of protons and neutrons; orbitals: areas where electrons are most likely to be found, organised by energy level; electrons: particles with negligible mass that are found in the orbitals surrounding the nucleus, adding negative charge; neutrons: a particle found in the nucleus with neutral charge holding protons together
- f. Atoms with the same number of protons but different numbers of electrons (same charge, different mass)
- g. C-14
- h. Ionic bonding is the bonding of atoms through electrostatic attraction, a result of one atom taking electrons from another to fill both their valence shells; covalent bonding is the bonding of atoms through the sharing of electrons.
- i. A chemical formula shows the number of each type of atom in a compound, and a structural formula creates a visual to show where the atoms are connected to each other in space.
- j. ion: an atom with a positive or negative net charge due to an unequal number of protons and electrons; cation: an ion with positive charge; anion: an ion with negative charge.
- k. Metals are characterised by high conductivity, lustre, and malleability. They lose electrons to nonmetals in ionic bonding. Nonmetals are found on the upper right side of the periodic table, and they take electrons during ionic bonding. They are

dull, brittle, and poor conductors of heat and electricity. They also form covalent bonds with each other. Metalloids are elements that fall in between these two sets of characteristics. They are semiconductors and can form both ionic and covalent bonds.

- l. Periods are the rows, and groups are the columns on the periodic table.
- m. The s-block, p-block, and d-block are the groups that the periodic table is divided into based on the highest energy electron orbital.
- n. Group 1, Group 2, Group 7, Group 8
- o.
 - i. Sodium chloride, calcium fluoride, lithium oxide, iron (III) oxide, silver (I) fluoride
 - ii. MgSe, RbI, Na₃P, Cu₂S, HgO
 - iii. Ammonium chloride, sodium hydroxide, silver nitrate, manganese (II) carbonate
 - iv. KBrO₃, Fe(NO₃)₃

6 Chapter 3

- a. imma bfr i have no clue
- b. imma bfr i have no clue 2.0
- c. An empirical formula shows the simplified ratio of atoms in a compound (for example, CH₂O) while a molecular formula shows the exact number of atoms in a molecule (C₆H₁₂O₆).
- d. Equations need to be balanced to perform calculations and to make sure that the law of conservation of mass is followed.
- e. aqueous (aq), solid (s), liquid (l), gas (g)
- f. The ratio of moles in one substance to moles of another in a balanced equation
- g. The reactant that limits the amount of product that can be made (and is used up completely)
- h. ima bfr i have NO CLUE. the last 5 questions gave me a confidence boost. now im on question 9 or 10 like “whats a stoichiometry”
- i. The theoretical amount of reactant that is expected to be produced
- j. The percent yield is the percent of the theoretical yield that was actually produced in the experiment. The formula is

$$\text{percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

k.

$$\begin{aligned}\frac{0.2 \text{ g}}{1 \text{ carat}} \times 2.5 \text{ carats} &= 0.5 \text{ g} \\ 0.5 \text{ g} \times \frac{1 \text{ mol}}{12.011 \text{ g}} &= 0.0416 \text{ mol} \\ 0.0416 \text{ mol} \times 6.022 \times 10^{23} \text{ atoms/mol} &\approx \boxed{2.507 \times 10^{22} \text{ atoms}}\end{aligned}$$

l. i.

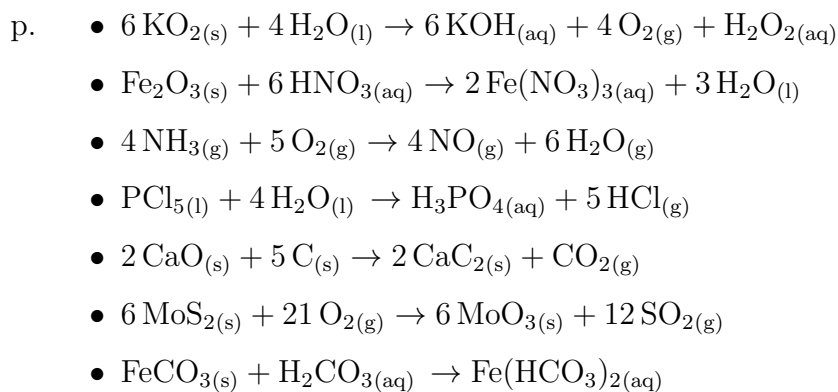
Molar mass of $\text{Fe}_2\text{O}_3 = 159.687 \text{ g/mol}$

$$150 \text{ g} = \frac{1 \text{ mol}}{159.687 \text{ g}} \approx \boxed{0.94 \text{ mol}}$$

m. uhh

n. uhh

o. uhh



q. uhh

r. uhh

7 Additional knowledge